Statistical Decision Making
45-752

Basics of Multiple Regression

The Four R's: Reading, Writing, Arithmetic, and Regression

This file is named: 03_slides.pptx
The accompanying Python notebook is: 03_notebook.ipnyb

For further reading on this topic see:
Chapter06_Introduction_to_Multiple_Regression.pdf

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Overview of Multiple Regression

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Multiple Regression

You covered some basics of multiple regression in the Prob-Stat course. We will build on the work you did in that course.

We will first discuss a description of an application by a Tepper alum to illustrate the usefulness of multiple regression.

Then we will lay out the general formulation of the multiple regression model.

We will view some graphs illustrating the multiple regression model with two right-hand-side variables.

We will develop diagnostics to assess the adequacy of the estimated model.

First, a recap of the essential rules for hypothesis testing.

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Review of Hypothesis Test Decision Rules

If your p-value is less than α, you:

- Reject the null hypothesis.

If your t statistic calculated using your data is bigger than the critical value $t_{\nu,\alpha/2}$ you:

- Reject the null hypothesis.

If your null hypothesis for a coefficient in not in the confidence interval, you:

- Reject the null hypothesis.

The above three decision rules always:

- Give the same result.

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Review of Hypothesis Testing Continued

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Multiple Regression: From A Tepper Alumnus

Application: Automobile manufacturers purchase thousands of tires each year. Tire acquisition is a major cost item.

A Tepper alumnus developed a tire-pricing model for use by his company. The objectives in developing the model were twofold:

1. Supply Chain Management: To provide a model that could be used to negotiate prices with suppliers.
2. Cost Estimation: To provide a model for designers, to estimate the cost of equipping a newly designed vehicle with tires.

The next slide provides more information about the data he collected. His company permitted him to share the data with us.

Today we will student a regression model with two right-hand-side variables to build understanding an intuition. In a later class, we will student the model developed by our alum.

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More Information About The Data from Our Alum

Data: There are 61 observations, one for each of 61 different types of tires
Variable to Be Predicted: Price (Dollars per Tire)

Potential predictor variables:
Diameter: Inside diameter of the tire in inches
Ratio: Ratio of height of tire sidewall to width of tread, in percent.
Variables denoting the speed safety classification of a tire:
- R rated for normal highway driving
- H rated to 90 miles per hour
- V rated to 120 miles per hour
- Z rated to 150 miles per hour

Type of exterior
- BW: Black Wall tire
- WW: White Wall tire
- OWL: Outside White Lettering

Terrain: equals 1 if all tire is for all Terrain vehicle, 0 otherwise
Weight: weight of tire in pounds
Tread: tread width in millimeters
Volume: The number of tires of this type purchased by the company in the reference year (thousands)

- We now develop the tools needed to conduct this analysis.

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Multiple Regression

The population multiple regression model is:

$$y_i = \beta_0 + \beta_1 x_{1i} + \beta_2 x_{2i} + \ldots + \beta_K x_{Ki} + \varepsilon_i$$

y is the dependent variable (e.g., tire price)

$x_1, x_2, \ldots x_K$ are independent variables (e.g., characteristics of tires)

ε_i unobserved error term

i=1,…,n denotes the observations (n=61 in the tire data)

- There are K+1 unknown coefficients in the model: $\beta_0, \beta_1, \beta_2, \ldots, \beta_K$
- Coefficient β_j is the change in y from a one-unit change in x_j when other x's are unchanged (e.g. change in tire price in response to a one inch increase in diameter of tire to be purchased).
- The estimated coefficients $b_0, b_1, \ldots, b_K$ are chosen to minimize the sum of squared errors.

Note: An optional appendix at the end of these slides provides more information about the derivation of the equations for estimating the coefficients.

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Ideal Conditions for Multiple Regression

The ideal conditions for application of multiple regression are the natural extension of those used in the simple regression model:

1. The ε_i are distributed independently of the x_{ik} for all observations, i=1,…,n, and all variables, k=1,…,K.

2. The ε_i are from a population with a mean of zero and a constant variance σ^2: $E(\varepsilon_i) = 0$, $Var(\varepsilon_i) = \sigma^2$.

3. The error terms are not correlated with each other: $E(\varepsilon_j \varepsilon_i)=0$ for all $i \neq j$.

Under the above conditions, the famous Gauss-Markov Theorem tells us that: Least squares is the Best Linear Unbiased Estimator of the coefficients in the multiple regression model. As a shorthand, least squares is BLUE.

- We'll build intuition as we go along, and we will see how to analyze data for situations in which these ideal conditions may not be satisfied.

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Two Right-Hand-Side Variables

We first study a model with two right-hand-side variables. I made up the data in this example. The next two slides are different views of the **same** regression plane and data.

- The plane is the estimated regression equation. Be sure you can point out the following:
  - The slope coefficient b_1
  - The slope coefficient b_2
  - The intercept b_0
  - What is the numerical value of the slope coefficient b_2? Hint: From the graph on the next page, you can see where the regression plane intersects the "back wall" of the diagram.
- One data point is plotted with a vertical line to the regression plane. For this point, be sure you can answer the following:
  - What are the numerical (x_1,x_2) values for this point?
  - Do you agree that the value of y for this point is greater than 1.2?
  - Where is the fitted value for this point?
  - What distance corresponds to the estimated error term?

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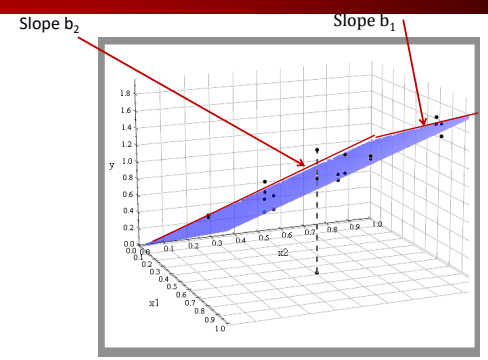
Regression Model with Two RHS Variables



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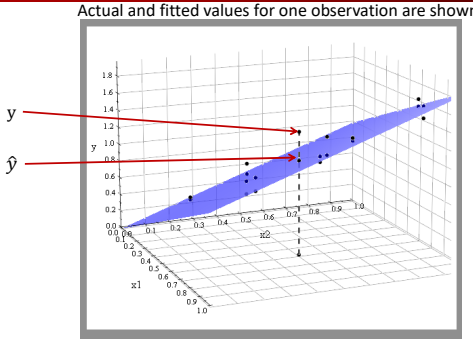
Actual Values, Fitted Values, & Residuals

Actual and fitted values for one observation are shown by arrows.



Predicted values:

$$\hat{y}_i = b_0 + b_1 x_{1i} + b_2 x_{2i}$$

Estimated Error Terms:

$$e_i = y_i - \hat{y}_i$$

What are the values of x_1 and x_2 for the observation with arrows?

Where on the figure is the error term for the observation with arrows?

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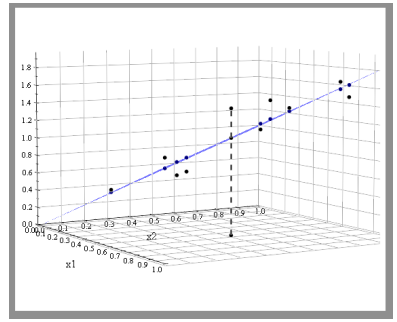
Two RHS Variables: Another View

This is another view of the same regression plane as previous page.



The error terms are the vertical differences between the actual and predicted values.

The estimated coefficients are chosen to minimize the sum of squared errors.

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Summary

- The geometric picture of the model with two right-hand-side variables is to help you visualize the result of a regression model with two right-hand variables.
- It will help you understand the model with many right-hand-side variables.

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Multiple Regression:
An Application and Interpretation

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Diamonds with Two Characteristics

- We now study a model with two right-hand side variables.
- We have data for 211 diamonds that vary in carat and clarity. All have the same cut and color.
- The file is Diamonds_211.xlsx.
- We will use regression analysis to study how price varies with carat and clarity.
- As you know, it is good practice to begin by getting a feel for the data.
- The following slide presents summary statistics for the 211 diamonds.

Diamonds with Two Characteristics

What is the standard deviation of diamond prices?

What is the weight of the largest diamond?

What is the range of values of Clarity?

We will investigate the salesperson variable later.

```
# view summary statistics
Diamonds_211.describe()
```

	Price	Carat	Clarity	Color	Cut	Salesperson
count	211.000000	211.000000	211.000000	211.0	211.0	211.000000
mean	8769.364929	1.239573	4.270142	11.0	2.0	5.208531
std	574[illegible]00127	0.511450	1.206336	0.0	0.0	2.469649
min	2056.000000	0.510000	2.000000	11.0	2.0	1.000000
25%	4037.500000	0.800000	3.000000	11.0	2.0	3.000000
50%	6914.000000	1.110000	4.000000	11.0	2.0	5.000000
75%	11750.500000	1.585000	5.000000	11.0	2.0	7.000000
max	28546.000000	2.470000	8.000000	11.0	2.0	9.000000

Diamonds Varying with Two Characteristics

We estimate our regression of Price against Carat and Clarity and name the result DiamReg:

```
# Estimate Diamond regression
DiamReg = tprstats.model("cs", "Price~Carat+Clarity", Diamonds_211)
```

We view the results using:

```
# view the regression summary
DiamReg.summary()
```

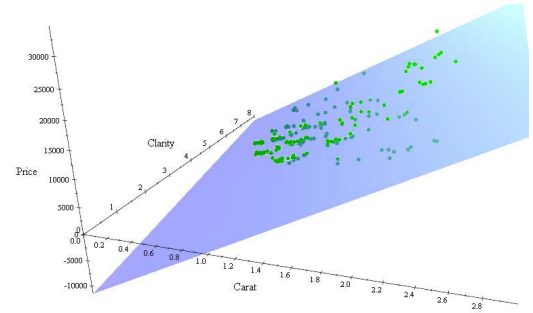
Dep. Variable:	Price	R-squared:	0.950
Model:	OLS	Adj. R-squared:	0.950
No. Observations:	211	F-statistic:	773.8
Covariance Type:	HC1	Prob (F-statistic):	4.54e-97

	coef	std err	t	P>|t|	[0.025	0.975]
Intercept	-1.16e+04	759.549	-15.278	0.000	-1.31e+04	-1.01e+04
Carat	1.193e+04	314.094	37.973	0.000	1.13e+04	1.25e+04
Clarity	1308.8177	124.664	10.499	0.000	1063.051	1554.584

A plot of the regression plane is shown on the next page.

An Optional Appendix at the end of this Note Set compares formulas for the Simple Regression Model and the Multiple Regression Model.

Diamonds Varying with Two Characteristics



Evaluating Your Regression

The "big three" items to consider in evaluating a regression coefficient:

Statistical Significance: Is the coefficient statistically significant?

Algebraic Sign: Does the algebraic sign of the coefficient make sense?

Magnitude: If the coefficient is significant, is the magnitude plausible? Is the estimated effect of the variable big enough to be important?

An additional important consideration:

Functional Form: Have you chosen the appropriate functional form for each of the variables in your model.

Note: If a coefficient is insignificant, it's algebraic sign and magnitude may not be very meaningful.

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Criteria for Specifying Your Regression

Include All Relevant Variables: Your knowledge of the problem you are studying will guide you in determining the relevant explanatory variables.

Choose Appropriate Functional Forms For Variables: We will study diagnostics for functional form and strategies for determining an appropriate functional form for each variable.

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RHS Variables – Interpreting Coefficients

Significance: From the estimated regression equation for our diamonds, we see that the p-values for Carat and Clarity are very small. Hence, both are statistically significant at the 5% level (and the 1% level).

Sign: Both coefficients have positive algebraic signs, as we would expect.

Magnitude: The coefficient of each right-hand-side variable tells us the predicted change in price from a one unit change in that right-hand-side variable while holding all other right-hand-side variables constant.

The interpretation of coefficients is straightforward:

- Our regression predicts that an increase in clarity by one unit would increase price by $1,309 holding carat weight constant.
- Our regression predicts that a one carat increase in weight holding clarity constant would increase price by $11,930.

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Useful Facts About Regression

When we estimate a regression:

1. The estimated error terms have a mean of zero.
   - Why?
   - If the estimated residuals did not have a mean of zero, we could change the intercept and reduce the sum of squared errors (SSE) of the regression.
2. The estimated error terms are uncorrelated with the explanatory variables.
   - Why?
   - If there is a positive correlation between the residuals and an explanatory variable, we could increase the coefficient on that variable and reduce the SSE.
   - If there is a negative correlation between the residuals and an explanatory variable, we could reduce the coefficient of that variable and reduce the SSE.
3. The estimated error terms are uncorrelated with the fitted values.
   - Why?
   - The fitted values are linearly related to the fitted values. Hence, this property follows from property 2.

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Interpretation Continued

We next discuss R^2 Statistics

Dep. Variable:	Price	**R-squared:**	0.950
Model:	OLS	**Adj. R-squared:**	0.950
No. Observations:	211	**F-statistic:**	773.8
Covariance Type:	HC1	**Prob (F-statistic):**	4.54e-97

	coef	std err	t	P>|t|	[0.025	0.975]
Intercept	-1.16e+04	759.549	-15.278	0.000	-1.31e+04	-1.01e+04
Carat	1.193e+04	314.094	37.973	0.000	1.13e+04	1.25e+04
Clarity	1308.8177	124.664	10.499	0.000	1063.051	1554.584

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R^2

In the model with one right-hand-side variable, we saw that the R^2 statistic is the proportion of variation in the left-hand-side variable that can be "explained" by variation in the right-hand-side variable.

- The same interpretation applies when there are many right-hand-side variables.
- Looking at the equation for diamonds, we see that R^2=.95. This tells us that 95% of the variation in the prices of these 211 diamonds can be attributed to variation in carat and clarity.
- The remaining 5% of price variation may be due to a host of unmeasured factors (e.g., subtle differences in the diamonds that aren't measured, skill in negotiation of buyer relative to seller, etc.)
- An optional appendix provides more detail about *R^2*.

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Interpreting R^2

The R^2 statistic is often over-emphasized, particularly by people who do not have a very deep understanding of regression.

- We would, of course, like for the R^2 to be high, but R^2 varies with the application. Hence, there is no "right" number for the R^2 statistic.
- The R^2 statistic does not tell us whether a model is a good or a bad model. A high R^2 is neither necessary nor sufficient for a model to be good.
- Sign, significance, and magnitude of coefficients are much more important.

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Adjusted R^2

The R^2 statistic always increases when another right-hand-side variable is added, even if the variable has only a slight effect in reducing the sum of squared errors.

- The adjusted R^2 imposes a penalty for adding a variable.
- If an additional variable yields only a small improvement in fit, the adjusted R^2 will go down when the variable is added.
- The difference between R^2 and adjusted R^2 typically makes little difference except in small samples.
- The adjusted R^2 statistic is slightly preferred to the standard R^2 statistic, but it is not a very important difference.
- You need to be familiar with it, however, because, in an interview or in a work situation, you may be asked about it when you are presenting your results.

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Email Message I Received From An Alum

In attempting to explain R-squared to a manager I got myself confused.

- I thought I had it straight in my mind, but trying to explain to another person did not go so well.
- A good question for future exams is to ask a question on interpreting R-squared.
- Never thought I'd be this deep in stats again. I kind of like it.

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Summary

- We saw how to apply the "big three" criteria (significance, sign, magnitude) to our model for diamonds that vary in carat and clarity.
- We saw how to calculate and plot the actual and fitted values, and how to plot the residuals.
- We also saw how to interpret two additional items, R^2 and R^2_{adj}

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Multiple Regression
Hypothesis Testing

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Regression Model for Diamond Data

We continue to analyze the regression model with 211 observations.
Key items we will discuss next are:

s_{b_j} Standard Errors of the Regression Coefficients

t-statistics

p-values

Confidence Intervals

	coef	std err	t	P>|t|	[0.025	0.975]
Intercept	-1.16e+04	759.549	-15.278	0.000	-1.31e+04	-1.01e+04
Carat	1.193e+04	314.094	37.973	0.000	1.13e+04	1.25e+04
Clarity	1308.8177	124.664	10.499	0.000	1063.051	1554.584

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Standard Errors of Coefficients

The Standard Error of a coefficient is a measure of the accuracy of the coefficient.

- The standard error of the coefficient of variable k is denoted s_{b_k}
- The standard error of a coefficient is used in calculating confidence intervals for individual coefficients.
- It is also used in the denominator of the t-statistic as shown on the next page.
- The larger a coefficient is relative to its standard error, the more accurate the coefficient is.
- The formula for calculating the standard error of a coefficient is rather messy and varies with the number of coefficients. The formula with two right-hand-side variables is in an optional appendix to these slides

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t-statistics in Multiple Regression

We construct the t-statistic for the coefficient of variable k as we did in the simple regression model:

$$t_{b_k} = \frac{b_k - \beta_{k,0}}{s_{b_k}}$$

β_k Unknown population coefficient of variable k
$\beta_{k,0}$ Null hypothesis for coefficient of variable k
b_k Estimated coefficient of variable k
s_{b_k} Estimated standard error of coefficient of variable k

- This formula, with $\beta_{k,0}$=0, is used to calculate each of the (K+1) entries in the column labeled "t" in the Python output.
- K is the number of right-hand side variables. In our diamond model, K=2.
- Including the coefficient of the intercept, we have K+1 coefficients.
- There are (n-K-1) degrees of freedom.
- Hence, in our diamond model with two right-hand-side variables, the t-statistics have (n-3) degrees of freedom.

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p-values

The column in the Python output labeled P>|t| reports the p-values for testing the null hypothesis that each corresponding population coefficient equals zero.

- The p-value for a coefficient provides the information you need for testing the null hypothesis that a population coefficient equals zero. If p-value<α, you reject the null hypothesis.
- The t-statistics are used to calculate the p-values, and they are used to calculate confidence intervals as we briefly review on the next slide.
- Often people report t-statistics instead of p-values, so you need to be comfortable using the t-statistic for hypothesis testing. You can always calculate your own p-value from a t-statistic using Python.
- Suppose you have a t-statistic=1.025 and 29 degrees of freedom. You can calculate the two-tailed p-value as follows:
  - In Python: 2*t.cdf(-1.025, 29)

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Confidence Intervals for Coefficients

The confidence interval for the coefficient of variable k is

$[b_k - t_{\nu,\alpha/2} * s_{b_k}, b_k + t_{\nu,\alpha/2} * s_{b_k}]$

Example: Calculate a 95% confidence interval for the coefficient of Clarity.

- The coefficient and std error are: $b_{clarity} = 1308.82$ and $s_{b_{clarity}} = 124.664$
- We have (211-3)=208 degrees of freedom.
- Hence, using Python command t.ppf(.975,208) we obtain: $t_{\nu,\alpha/2}$ = 1.97
- Our confidence interval is: [(1308.82-1.97*124.664), (1308.82+1.97*124.664)] = [1063.2, 1554,4]

We calculate that the confidence interval for Clarity is approximately [1063.2, 1554.4]
As you know, Python does the work for us and reports confidence intervals.

	coef	std err	t	P>|t|	[0.025	0.975]
Intercept	-1.16e+04	759.549	-15.278	0.000	-1.31e+04	-1.01e+04
Carat	1.193e+04	314.094	37.973	0.000	1.13e+04	1.25e+04
Clarity	1308.8177	124.664	10.499	0.000	1063.051	1554.584

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Hypothesis Tests of Coefficients

Suppose we wish to test the null hypothesis that $\beta_{carat,0}$ = 13,000 using a 5% significance level.

If you have the data, you can run the regression and calculate the confidence interval for the coefficient:

- We see that the null hypothesis value β_{carat} = 13,000 is not in the 95% confidence interval [11300, 12500] on the previous page.
- Hence we do reject the null hypothesis at the 95% level.

Often, you will not have the data when you are reading a study using regression analysis, so you need to be able to do such tests on your own using the coefficient and std error, as follows:

- The form of the t-statistic: (EstimatedCoef - Null)/StdErrorCoef
- The relevant t-statistic is then (11927-13000)/314.1 =-3.42
- There are 211 observations, so degrees of freedom=208
- The p-value is obtained using: 2*t.cdf(-3.42, 208)
- The result is p-value =.00075, so we reject this null hypothesis.

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A Handy Way to Test a Hypothesis

When you have the data, there is another way to test the hypothesis on the previous slide.

Issue the following tprstats command:

DiamReg.wald_test("Carat = 13000")

- You obtain p-value = .0008
- On the previous slide, we obtained .00075.
- Up to roundouf error, the above two values are equal as they should be.
- Note: The above command is called the Wald test after it's inventor Abraham Wald. We will often use Wald as a shorthand to refer to the above test.
- This test will quickly become one of your favorites.

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Versatility of the Wald Test

We can test a hypothesis about more than one coefficient using the Wald test.

```
# Test the null hypothesis that the population coefficient of Carat = 12,000
# and the population coefficient of Clarity =1250.
DiamReg.wald_test(('Carat = 12000, Clarity=1250'))
```

```
p-value:  0.7698
```

- We obtain p-value= .7698.
- This is greater than .05. Hence, we do not reject the null hypothesis.

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Summary

We discussed:

- Computation and use of standard errors of the coefficients
- t-statistics in the multiple regression case have (n-K-1) d.f.
- Hypothesis testing for individual coefficients
- Confidence intervals for individual coefficients
- Use of the Wald test

We illustrated with an application to diamond data

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Multiple Regression Prediction

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Diamonds Varying with Two Characteristics

We continue to use the regression for our diamond data.

	coef	std err	t	P>|t|	[0.025	0.975]
Intercept	-1.16e+04	759.549	-15.278	0.000	-1.31e+04	-1.01e+04
Carat	1.193e+04	314.094	37.973	0.000	1.13e+04	1.25e+04
Clarity	1308.8177	124.664	10.499	0.000	1063.051	1554.584

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Prediction

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Prediction and Prediction Intervals

A jeweler has been offered the opportunity to purchase three new diamonds. All three diamonds have the same color and cut as the diamonds in our data set (Color=11 and Cut=2). The three diamonds have carat and clarity as follows:

CARAT	CLARITY
0.80	7
1.30	6
2.50	2

The jeweler can buy the first diamond for $8,500, the second for $12,000, and the third for $18,000. The jeweler is considering whether to buy any of them.

- How might we use our model to help with this decision?
- For what prices could the jeweler expect to sell these diamonds.
- Using our regression equation, let's predict the price that the jeweler would obtain for each and form a 95% prediction interval for each.

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Solution to Our Prediction Question

Create new variables in Colab with the values to be used for prediction as shown below.

```
# Put the Carat and Clarity data into a data frame named Xnew.
Xnew = pandas.DataFrame({
    'Carat': [0.8, 1.3, 2.5],
    'Clarity': [7,6,2]
})
```

Then issue the command below. Note that the command begins with the name of our regression.

```
# predict intervals
DiamReg.prediction_intervals(Xnew)
```

We obtain the following:

	Predicted	Lower	Upper
0	7099.394498	4490.648361	9708.140634
1	11754.149530	9164.138653	14344.160407
2	20831.453346	18177.742958	23485.163733

Now, look back at the price offers on the previous slide. Any bargains there?

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Interpretation of Prediction Results

The jeweler can buy the first diamond for $8,500. The predicted price at which the jeweler could sell the diamond is $7,099. Hence, this is no bargain.

The jeweler can buy the second diamond for $12,000. The predicted price at which the jeweler could sell the diamond is $11,754. Hence, this is also no bargain.

The jeweler can buy the third diamond for $18,000. The predicted price at which the jeweler could sell the diamond is $20,831. The 95% prediction interval is [$18250, $23412].

- Note that the purchase price is slightly below the lower bound of the prediction interval for the price at which the jeweler can sell the diamond. Hence, the predicted probability of a sale price less than the purchase price is a bit less than 2.5% whereas the upper bound of the interval is far above the purchase price.
- Thus, this diamond looks like a profitable purchase for the jeweler.

The notebook accompanying this note set shows you how to calculate the predicted prices and prediction intervals for all 211 diamonds in our original dataset.

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Another Way to Implement The Prediction

An alternative approach to do the same prediction is to enter the data for your prediction into an Excel file.

I put the data, as shown below, into a file I named Diamond_Prediction_Data.xlsx. It is important to put the names in the first row exactly as they appear in your regression (including matching upper and lower case).

	A	B
1	Carat	Clarity
2	0.8	7
3	1.3	6
4	2.5	2

Then, upload the Excel file into the data sets in Colab and run the following commands.

```
# Import the file
Diamond_Prediction_Data = pandas.read_excel("data/Diamond_Prediction_Data.xlsx")
# Execute the following command.
DiamReg.prediction_intervals(Diamond_Prediction_Data)
```

Putting predictors in a spreadsheet is quite useful if you have many new cases for which you wish to do predictions.

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Summary of Multiple Regression Thus Far

Prediction is one of the key uses of multiple regression.

We studied how to do prediction and calculate prediction intervals.

We studied a managerial application, using prediction to guide a decision about purchase of diamonds.

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Statistical Decision Making
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Multiple Regression:
Bias Caused by Leaving Out A Variable That Should Be Included

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Diamonds Varying with Two Characteristics

Our regression of Price against Carat and Clarity is shown below.

OLS Regression Results

Dep. Variable:	Price	**R-squared:**	0.950
Model:	OLS	**Adj. R-squared:**	0.950
No. Observations:	211	**F-statistic:**	773.8
Covariance Type:	HC1	**Prob (F-statistic):**	4.54e-97

	coef	std err	t	P>|t|	[0.025	0.975]
Intercept	-1.16e+04	759.549	-15.278	0.000	-1.31e+04	-1.01e+04
Carat	1.193e+04	314.094	37.973	0.000	1.13e+04	1.25e+04
Clarity	1308.8177	124.664	10.499	0.000	1063.051	1554.584

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RHS Variables – Interpreting Coefficients

Suppose we mistakenly leave out a variable from our multiple regression model?

Does this have any effect on coefficients or significance levels of other variables?

Let's consider the diamond data. Suppose we were to leave out the carat variable.

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RHS Variables and Variable Exclusion

Consider the coefficient of Clarity when Carat is not included.

	coef	std err	t	P>|t|	[0.025	0.975]
Intercept	1.253e+04	1123.542	11.155	0.000	1.03e+04	1.47e+04
Clarity	-881.2864	278.917	-3.160	0.002	-1431.138	-331.435

The coefficient of Clarity is negative and significant!

We see that the **bias** can be great if we leave out a variable that belongs.

- Can you provide an intuitive explanation for why the algebraic sign of Clarity is the opposite of what we would expect?
- Hint: Do large or small diamonds tend to have the highest clarity?

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Effect of Leaving Out a Variable that Should be Included

Let's calculate the correlation of Carat and Clarity.

```
# Calculate Correlation of Carat and Clarity
Diamonds_211['Carat'].corr(Diamonds_211['Clarity'])
```

- You will find the correlation equals -.43
- This tells us that large diamonds tend to have more imperfections (lower clarity) than small ones.
- This is intuitive: It is easier to cut a small diamond to avoid imperfections than to do so for a large diamond.

But, despite the greater number of imperfections, large diamonds sell for more than small ones.
⇒ It follows that diamonds with more imperfections (low clarity) sell for more than diamonds with few imperfections (high clarity).

Therefore, if we fail to control for size of the diamond, we obtain the spurious result that higher clarity reduces the price of a diamond.
The bias of the coefficient of clarity is very large when we leave out a variable that should be in the regression model (carat).

This is fundamental and intuitive. Leaving out a variable that should be in the regression can bias the coefficients of other variables in the regression.

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Another Example: Predicting Tepper School Graduating GPA's

Dependent variable is overall GPA at graduation
When GMATVERBAL is included, GMATWRITING has small insignificant coefficient.
When GMATVERBAL is excluded, GMATWRITING has much larger and significant coefficient.
What is the intuition?

	coef	std err	t	P>|t|
Intercept	1.9187	0.138	13.941	0.000
GMATQUANT	0.0161	0.002	7.621	0.000
GMATVERBAL	0.0123	0.002	6.330	0.000
GMATWRITING	0.0080	0.011	0.739	0.461
UGPA	0.1466	0.019	7.534	0.000
WORKYEARS	0.0261	0.009	2.801	0.005
I(WORKYEARS ** 2)	-0.0021	0.001	-2.805	0.005

	coef	std err	t	P>|t|
Intercept	2.4417	0.131	18.613	0.000
GMATQuant	0.0120	0.002	6.182	0.000
GMATWriting	0.0366	0.011	3.447	0.001
UGPA	0.1473	0.020	7.446	0.000
Workyears	0.0203	0.009	2.368	0.018
I(Workyears ** 2)	-0.0018	0.001	-2.445	0.015

For reasons of confidentiality, I have not shared these data with you.

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Interpretation of Previous Slide

Interpretation of result on previous slide

- Verbal and writing skills are positively correlated.
- When we include GMATVERBAL, it captures the effect of verbal skills, and GMATWRITING contributes little beyond GMATVERBAL.
- When we leave out GMATVERBAL, the GMATWRITING variable attempts to do double duty. It captures the effect of writing ability and picks up part of the effect of other verbal skills because of the correlation of writing and verbal skills.
- Hence, if we mistakenly leave out GMATVERBAL, we obtain an estimated coefficient for GMATWRITING that is biased upward.

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Summary: Effects of Leaving Out a Variable

- Leaving a variable out of a regression that belongs in the regression can bias the coefficients of variables that remain in the regression.
- To obtain unbiased coefficient estimates, we need to include other variables that belong in the model, even if we are not particularly interested in the coefficients of those variables.
- As a manager, you need to rely on your knowledge of the problem you are studying to be sure that you include variables that should be in your model.

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Diagnostics to Assess Adequacy of Model

- When estimating a regression, it is important to assess whether the model adequately captures the relationships in the data.
- We next calculate fitted values and residuals.
- We then plot graphs that are informative about model fit and functional form.

Fitted Values and Residuals

Using our regression results, we can calculate the fitted values of the dependent variable. Recall that the fitted value for observation i is :

$$\hat{y}_i = b_0 + b_1 x_{1i} + b_2 x_{2i} + \ldots + b_K x_{Ki}$$

We can also obtain the estimated errors (residuals): $e_i = y_i - \hat{y}_i$

The following Python commands calculate the fitted values and residuals from our diamond regression and names them yhat and resid respectively:

```
# Calculate fitted values and name them yhat
yhat = DiamReg.predict()
# Calculate residuals and name them resid
resid = DiamReg.resid
```


Placing Actual Values, Fitted Values and Residuals in a Spreadsheet

We can put the actual prices, fitted values, and residuals into a data frame (a spreadsheet). The following creates a data frame named ActFitResid that contains Price, yhat, and resid.

```
# Create a data frame containing actual prices, fitted prices, and residuals
ActFitResid= pandas.DataFrame({
    "Price": Diamonds_211['Price'].copy(),
    "yhat": yhat,
    "resid": resid
    })
ActFitResid
```

The results are displayed in Colab by the last line of the above.
If we want to save the results, we can write the results to an Excel file as follows:

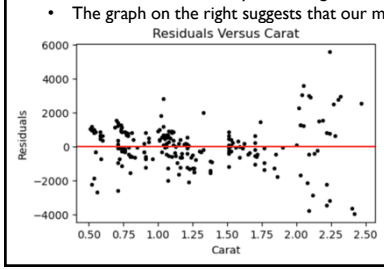
```
# If you wish to save your results, you can place the results to an Excel file
# The file can then be put the in your Colab drive with your notebooks and data.
ActFitResid.to_excel('/content/drive/MyDrive/ActFitResid.xlsx', index=False)
```

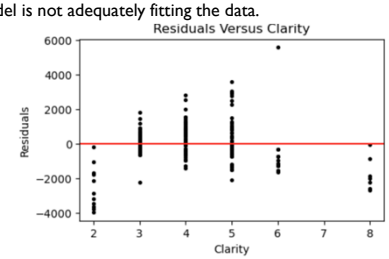
The ActFitResid.Excel file will be placed in the same folder on your computer as your Colab notebook.

Plot Residuals Versus Explanatory Variables

It is often informative to plot residuals versus explanatory variables.

- The graph on the left below is a plot of residuals against Carat.
  - What word describes the pattern we see in this plot? Have we dealt this with in our estimation?
- The graph on the right below shows a nonlinear pattern of the residuals versus clarity.
  - The residuals are negative for Clarity=2 and Clarity=8.
  - The residuals for Clarity=3, Clarity=4, and Clarity=5 have more positive than negative values.
  - The residuals for Clarity=6 are negative except for one large outlier. No diamonds have Clarity=7.
  - The graph on the right suggests that our model is not adequately fitting the data.



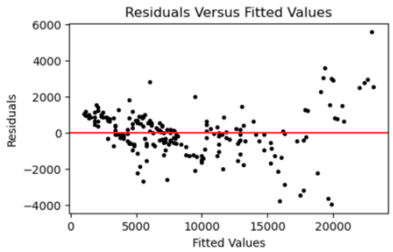


Plot Residuals Versus Fitted Values

It is also informative to plot residuals against fitted values.

- Near the middle, the residuals are disproportionately negative. Near the ends, the residuals are disproportionately positive.
- As explained on the next slide, this is more evidence that our model has not fully captured the relationship in the data.

See the Python notebook for commands to construct this plot.



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Implications of Graph on Previous Slide

Let's work through the interpretation of the graph on the previous slide.

- yhat is calculated from our estimated regression.
- Hence, all variation in yhat is due to variation in the explanatory variables in our model.

If the model fully captures the effects of the explanatory variables, there will be no systematic relationship between the residuals and the fitted values.

- Why? Systematic variation between the residuals and the fitted values implies systematic variation between the residuals and the explanatory variables.

A nonlinear relationship between the residuals and yhat means that we have not fully captured the relationship of the dependent variable to the explanatory variables.

Don't worry if you don't fully grasp this now. We will cover choice of functional form in detail in the near future. We will also introduce a test for functional form called the Ramsey Test that is closely related to the graph on the previous slide.

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Interpreting Graph of Actual, Predicted, and Prediction Intervals

In the data file, diamonds are ordered as shown on the right.

- Diamonds are ordered by Clarity, and then by Carat for each Clarity.

The graph of actual prices, predicted prices, and prediction intervals are shown on the next slide.

- The dots are the actual prices. The red curves are the predicted prices. The blue curves are 95% prediction intervals.
- For Clarity 2, several black dots are below the lower blue curve. For Clarity 4, several are above the blue curve.

In the next slide deck, we will build a model that better fits the data.

1	Price	Carat	Clarity	Color	Cut	Salesperson
2	11102	1.7	2	11	2	4
3	10228	1.7	2	11	2	3
4	11389	1.86	2	11	2	3
5	13504	2.03	2	11	2	3
6	13657	2.08	2	11	2	3
7	12166	2.09	2	11	2	8
8	13394	2.12	2	11	2	1
9	14025	2.22	2	11	2	8
10	14515	2.24	2	11	2	9
11	15966	2.4	2	11	2	8
12	15890	2.42	2	11	2	8
13	6312	1.02	3	11	2	4
14	5894	1.04	3	11	2	7
15	5632	1.06	3	11	2	9
16	5632	1.06	3	11	2	7
17	5626	1.07	3	11	2	6
18	5746	1.07	3	11	2	6
19	6121	1.08	3	11	2	4
20	5791	1.09	3	11	2	1

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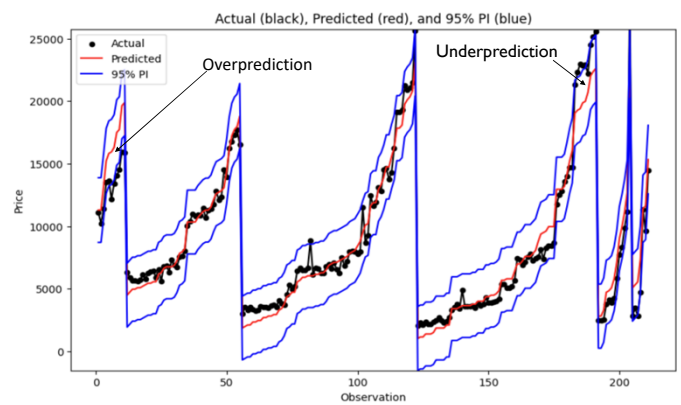
Graph of Actual, Predicted, and Prediction Interval



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Summary

In this slide deck, we have defined the multiple regression model and have shown how to estimate a model of prices for diamonds that vary in carat and clarity.

We covered interpretation of coefficients and hypothesis testing.

We developed diagnostics that can be used to assess adequacy of the estimated model.

These diagnostics show that there are shortcomings to the existing model.

In our next slide deck, we will cover functional form and develop an improved model for these data.

Appendix: Least Squares Estimation

As with the simple model with one right-hand-side variable, we to choose estimators (b_j) of the coefficients to minimize sum of squared estimated errors. The estimated model will be of the form:

$y_i = b_0 + b_1x_{1i} + b_2x_{2i} + \ldots + b_kx_{Ki} + e_i$

The estimated error, e_i, for observation i is obtained by solving for e_i:

$e_i = (y_i - b_0 - b_1x_{1i} - b_2x_{2i} - \ldots - b_kx_{Ki})$

The sum of squared errors is then:

$$\sum_{i=1}^{n} e_i^2 = \sum_{i=1}^{n} \left(y_i - b_0 - b_1x_{1i} - b_2x_{2i} - \ldots - b_Kx_{Ki} \right)^2$$

Differentiating with respect to $b_0, b_1, \ldots, b_K$ and setting the resulting equations equal to zero yields K+1 linear equations that can be solved for the K+1 coefficients.

Appendix: Interpreting Std Errors of Coefficients

With two right-hand-side variables, the estimated Std Error of coef b_1 is:

$$s_{b_1} = \sqrt{\frac{s_e^2}{\sum (x_1 - \bar{x}_1)^2 (1 - r_{x_1x_2}^2)}}$$

In this formula, $r_{x_1x_2}$ is the simple correlation between x_1 and x_2.

We want the Std Error of the coef to be small.
Is there anything we can do to make the Std Error smaller?

- It helps if the variance of the errors is small, but we can't control that.
- It helps if the correlation of x_1 and x_2 is small, but we can't control that either unless we are doing an experiment.
- It helps if $\sum (x_i - \bar{x})^2$ is large. We can make this larger if we can get a larger sample because this sum of squares gets bigger as more terms are added.

Bottom Line: If you can increase the size of your sample, you can get more precise estimates of the coefficients.

Appendix: Useful Formulas

Formulas:

Fitted values: $\hat{y}_i = b_0 + b_1x_{1i} + b_2x_{2i} + \ldots + b_Kx_{Ki}$

Estimated errors: $e_i = y_i - \hat{y}_i$

Total sum of squares: $SST = \sum_{i=1}^{n} (y_i - \bar{y})^2$

Regression sum of squares: $SSR = \sum_{i=1}^{n} (\hat{y}_i - \bar{y})^2$

Sum of squared errors: $SSE = \sum_{i=1}^{n} e_i^2$

$$R^2 = \frac{SSR}{SST} = \frac{SST - SSE}{SST} = 1 - \frac{SSE}{SST} = 1 - \frac{\sum_{i=1}^{n} e_i^2}{\sum_{i=1}^{n} (y_i - \bar{y})^2}$$

$$R^2_{adj} = 1 - \frac{\left(\frac{SSE}{n-K-1}\right)}{\left(\frac{SST}{n-1}\right)}$$

t-statistic for coefficient j: $t_{b_j} = \frac{b_j - \beta_j}{s_{bj}}$

Appendix: Detail Regarding R^2

As in the simple regression, the total sum of squares of y is: $\sum_{i=1}^{n}(y_i-\bar{y})^2$.

This can be partitioned into the portion "explained" by the regression, $\sum_{i=1}^{n}(\hat{y}_i-\bar{y})^2$, and the "unexplained" portion, the sum of squared errors, $\sum_{i=1}^{n}e_i^2$

The R^2 statistic is the "explained" over the "total"
R^2 is a fraction between 0 and 1.

$$R^2=\frac{\sum_{i=1}^{n}(\hat{y}_i-\bar{y})^2}{\sum_{i=1}^{n}(y_i-\bar{y})^2}$$

The R^2 statistic is the proportion of variation in the left-hand-side variable that can be "explained" by variation in the right-hand-side variables.

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Appendix: Detail for Adjusted R^2_{adj}

As shown on the previous slide, the R^2 statistic is the ratio of the "explained variance" over the total.
This can be written equivalently as 1 minus the ratio of the "unexplained" over the total.
The unexplained variance is: $\sum_{i=1}^{n}(e_i)^2$

$$\underbrace{R^2=\frac{\sum_{i=1}^{n}(\hat{y}_i-\bar{y})^2}{\sum_{i=1}^{n}(y_i-\bar{y})^2}}_{\text{Explained over total}} = \underbrace{1-\frac{\sum_{i=1}^{n}(e_i)^2}{\sum_{i=1}^{n}(y_i-\bar{y})^2}}_{\text{1-(unexplained/total)}}$$

Adjusted R^2_{adj} adjusts for the number of coefficients, K, in the model. This introduces a small penalty for adding a variable. If you study the formula a bit, you will see that if K goes up and the sum of squared errors does not go down, the R^2_{adj} will go down:

$$R^2_{adj}=1-\frac{\sum_{i=1}^{n}(e_i)^2/(n-K-1)}{\sum_{i=1}^{n}(y_i-\bar{y})^2/(n-1)}$$

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Appendix: Comparing b_1 in Simple vs Multiple Regression

Coefficient of x_1 from simple regression of x_1 on y:

$$b_1=r_{x,y}\frac{s_y}{s_{x_1}}$$

Coefficient of x_1 in a multiple regression that also includes x_2:

$$b_1=\frac{s_y(r_{x_1,y}-r_{x_1x_2}r_{x_2y})}{s_{x_1}(1-r^2_{x_1x_2})}$$

Question: What special condition must hold in the data for the above two estimators to be the same?
Answer: x_1 and x_2 must be uncorrelated (i.e $r_{x_1x_2}=0$).

If we fail to include x_2 when it belongs in the equation, the coefficient of x_1 will generally be biased. The direction of the bias can be either upward or downward.
Thus: **Excluding a variable that belongs in a model can cause bias in the coefficients of variables that remain in the model.**

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